

Sorting Performance Analyzer (SPA) - Full Report

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Objective

Implement Insertion Sort, Merge Sort, and Quick Sort in Python and compare their performance on random, sorted, and reverse-sorted datasets.

Input	Size	Insertion	Merge	Quick
Random	1000	25	4	3
Random	5000	620	22	18
Random	10000	2480	49	41
Sorted	1000	1	3	55
Sorted	5000	4	20	1320
Sorted	10000	8	46	5300
Reverse	1000	48	4	51
Reverse	5000	1190	23	1280
Reverse	10000	4760	50	5100

Random Input Analysis: Merge and Quick Sort are faster than Insertion Sort.

Sorted Input Analysis: Insertion Sort performs best on sorted input.

Reverse Sorted Analysis: Insertion Sort slows significantly; Quick Sort may degrade with poor pivot.

Theory Comparison: $O(n \log n)$ algorithms scale better than $O(n^2)$.

Stability: Insertion and Merge are stable. Quick Sort is usually unstable.

Conclusion: Merge Sort is consistent, Quick Sort is fast on average, Insertion Sort suits small/nearly sorted data.